

NSU BSc CSE — Full Course List

Fahim · 134 credit-hour program · MAT112 & ENG102 waived

Note: descriptions below summarize standard content for each course title. Titles, prerequisites, and credit values were confirmed against NSU's published curriculum earlier in this project; exact catalog wording for descriptions could not be re-verified live due to a search outage — cross-check against your official handbook once enrolled.

General Education & English

ENG103 — Intermediate Composition (3 cr)

Prerequisite: ENG102 (waived)

Builds paragraph- and essay-level academic writing, grammar, and structure.

ENG105 — Advanced Writing Skills (3 cr)

Prerequisite: ENG103

Argumentative and research-based writing, source use, citation, and revision.

ENV107 — Environmental Science (3 cr)

Prerequisite: None

Introduces ecological systems, sustainability, and environmental issues.

GED Elective — General Education (choose 3) (9 cr)

Prerequisite: None

Selected from NSU's approved humanities/social science/arts list; options vary by term.

Mathematics & Basic Sciences

MAT116 — Pre-Calculus (3 cr)

Prerequisite: High school math (MAT112 waived)

Functions, graphs, trigonometry, and exponential/logarithmic functions, preparing for calculus.

MAT120 — Calculus & Analytic Geometry I (3 cr)

Prerequisite: MAT116

Limits, derivatives, and their applications.

MAT125 — Introduction to Linear Algebra (3 cr)

Prerequisite: MAT116

Vectors, matrices, systems of linear equations, and eigenvalues.

MAT130 — Calculus & Analytic Geometry II (3 cr)

Prerequisite: MAT120

Integration techniques, sequences, and series.

MAT240 — Calculus & Analytic Geometry III (3 cr)

Prerequisite: MAT130

Multivariable calculus and partial derivatives.

MAT250 — Calculus & Analytic Geometry IV (3 cr)

Prerequisite: MAT240

Vector calculus, multiple integrals, and line/surface integrals.

MAT350 — Engineering Mathematics (3 cr)

Prerequisite: MAT250

Differential equations and their applications in engineering problems.

MAT361 — Probability and Statistics (3 cr)

Prerequisite: MAT250

Probability theory, distributions, and statistical inference.

PHY107 + 107L — General Physics I (+ Lab) (3 cr)

Prerequisite: MAT120

Mechanics, motion, and energy, with a hands-on lab component.

PHY108 + 108L — General Physics II (+ Lab) (3 cr)

Prerequisite: MAT240, PHY107

Electricity, magnetism, and waves, with a hands-on lab component.

CHE101 — General Chemistry (3 cr)

Prerequisite: High school chemistry

Foundational chemistry concepts relevant to engineering coursework.

Core CSE Courses**CSE115 + 115L — Computing Concepts (+ Lab) (4 cr)**

Prerequisite: None

Introduction to computers, problem-solving, and basic programming logic.

CSE135 + 135L — Fundamentals of Programming (+ Lab) (4 cr)

Prerequisite: CSE115, MAT120

Structured programming: variables, control flow, functions, and arrays.

CSE173 — Discrete Mathematics (3 cr)

Prerequisite: CSE135, CSE225

Logic, set theory, combinatorics, and graph theory foundational to CS theory.

CSE225 + 225L — Data Structures (+ Lab) (4 cr)

Prerequisite: CSE135

Stacks, queues, trees, linked lists, and hashing, with performance analysis.

CSE231 — Digital Logic Design (3 cr)

Prerequisite: CSE225, CSE173

Boolean algebra, logic gates, and combinational/sequential circuit design.

CSE232 — Computer Organization and Design (3 cr)

Prerequisite: CSE231

CPU architecture, memory hierarchy, and instruction sets.

CSE243 + 243L — Electrical Circuits (+ Lab) (4 cr)

Prerequisite: MAT120

Circuit analysis, components, and lab measurement techniques.

CSE253 + 253L — Electronics I (+ Lab) (4 cr)

Prerequisite: CSE243

Diodes, transistors, and analog circuit behavior.

CSC273 — Introduction to Theory of Computation (3 cr)

Prerequisite: CSE173, CSE225

Automata theory, formal languages, and computability.

CSE257 — Numerical Methods (3 cr)

Prerequisite: CSE225, MAT125, MAT240

Computational techniques for solving mathematical problems numerically.

CSE311 — Database Systems (3 cr)

Prerequisite: CSE225

Relational model, SQL, normalization, and database design.

CSE323 — Operating Systems Design (3 cr)

Prerequisite: CSE225, CSE232

Processes, memory management, scheduling, and concurrency.

CSE326 — Compiler Construction (3 cr)

Prerequisite: CSE232, CSC273

Lexical analysis, parsing, and code generation.

CSE327 — Software Engineering (3 cr)

Prerequisite: CSE225

Software development lifecycle, requirements, design, and testing methodologies.

CSE331 — Microprocessors and Assembly Language (3 cr)

Prerequisite: CSE232

Assembly programming and microprocessor architecture.

CSE338 — Computer Networks (3 cr)

Prerequisite: CSE232, MAT361

Networking protocols, the OSI/TCP-IP model, and data transmission.

CSE373 — Design and Analysis of Algorithms (3 cr)

Prerequisite: CSC273, MAT361

Algorithm design paradigms, complexity analysis, and optimization.

CSE413 — Digital Electronics & Microprocessor Lab (2 cr)

Prerequisite: CSE331

Applied lab work pairing digital logic design with microprocessor programming.

CSE482 — Internet and Web Technology (3 cr)

Prerequisite: CSE338

Web architecture, client-server design, and internet application development.

Specialization Trail (choose 3 courses from one trail)

Trail note — Computer Networks & Systems trail recommended (9 cr)

Prerequisite: Varies by course

Given the cybersecurity direction: CSE433, CSE438, CSE439, CSE463, and CSE487 are the trail options most aligned with that goal. Exact descriptions and prerequisites should be confirmed with your advisor when trail selection opens, since the search outage prevented re-verifying these live.

Open Electives

Open Elective — Any 2 approved open electives (6 cr)

Prerequisite: Varies by course

Chosen freely from NSU's approved elective list, subject to advisor approval.

Capstone

CSE499 — Project (3 cr)

Prerequisite: 100+ credits completed

Capstone project applying CSE coursework to a substantial independent or team project.